Practice workbook · Describe how to carry out a food test, including DCPIP for vitamin C

Writing the method for Benedict's, iodine, biuret, emulsion or DCPIP tests on foods, urine, plasma or tubing contents, including safe conduct and a DCPIP comparison of two juices.

17 question parts: 1 worked-example retry and 16 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

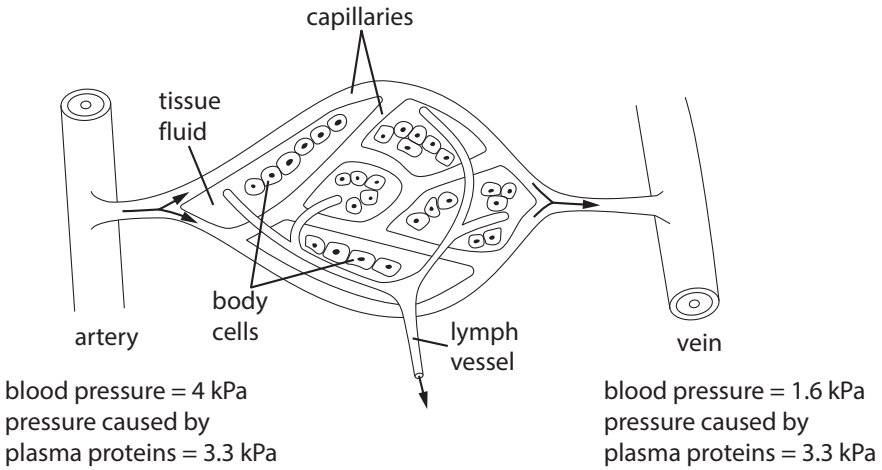
Type ID: HBI-FOOD-TEST-PROCEDURE

1. Question 7 · part b · 4 marks

June 2025 Paper 1

Worked-example retry · Attempt part b only.

7 The diagram shows a capillary network.



(a) (i) Explain the change in blood pressure from the artery to the vein.

(2)

(ii) Explain why the pressure caused by the plasma proteins does not change.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

1. Question 7 · part b · 4 marks

June 2025 Paper 1

Worked-example retry · Attempt part b only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(b) Describe how a sample of plasma could be tested for the presence of proteins.

(4)

(c) Tissue fluid is formed by the movement of molecules from the blood. This occurs in the capillary network.

Use data from the diagram to determine the direction of movement of fluid between the capillaries and tissue fluid at the artery end and at the vein end of the capillary network.

(4)

OLevels.com SCHOOL

1. Question 7 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 1 · part c · 3 marks

January 2020 Paper 1

Further practice · Attempt part c only.

1 (a) Food packaging gives information about the food it contains.

The photograph shows some of the information given on a package containing yogurt.



(Source: © studiomode / Alamy Stock Photo)

The instructions state that the yogurt should be stored in a fridge at 4°C.

Why should this instruction be followed?

(1)

- ☒ **A** to make the yogurt taste better
- ☒ **B** to kill all bacteria in the yogurt
- ☒ **C** to slow down the growth of bacteria in the yogurt
- ☒ **D** to improve the appearance of the yogurt

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

2. Question 1 · part c · 3 marks

January 2020 Paper 1

Further practice · Attempt part c only.

(b) The diagram shows the nutritional label on a package containing yogurt.

Nutrition Facts

Serving Size
1 container (150g)
Energy in kJ 480
Energy from Fat 0

Amount/Serving	% Daily Value*	Amount/Serving	% Daily Value*
Total Fat	a%	Sodium	2%
Saturated Fat	0%	Total Carb.	6%
Trans Fat		Sugars	
Cholesterol	1%	Protein	22%

Vitamin A 4% • Calcium 10% • Vitamin D 20%

Not a significant source of dietary fibre, vitamin C and iron.

(i) State which nutrient on the label is required for growth and repair of body tissue.

(1)

.....

(ii) Why is this yogurt suitable for a person who wants to reduce their body mass?

(1)

- ☒ **A** it contains vitamins C and D
- ☒ **B** it contains less than 5 mg of cholesterol
- ☒ **C** it does not contain energy from carbohydrates
- ☒ **D** it does not contain energy from fat

(c) Yogurt contains reducing sugars.

Describe a method to test for reducing sugars.

(3)

.....

.....

.....

.....

.....

.....

(Total for Question 1 = 6 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

2. Question 1 · part c · 3 marks · Working and answer

Working and answer

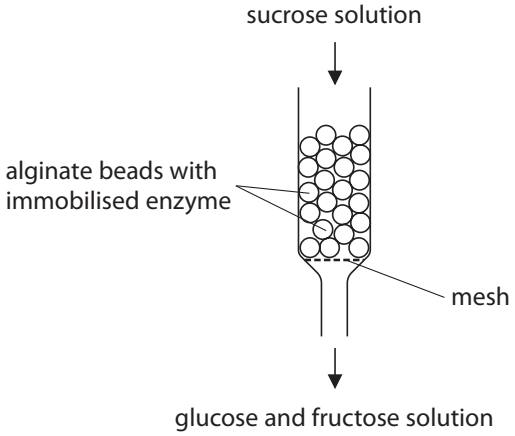
OLevels.com SCHOOL

3. Question 6 · part a.ii · 3 marks

November 2020 Paper 1

Further practice · Attempt part a.ii only.

6 (a) The diagram shows how immobilised enzymes can be used to convert sucrose into glucose and fructose.



(i) Name the immobilised enzyme used in this process.

(1)

.....

(ii) One advantage of using immobilised enzymes is that they do not contaminate the products, glucose and fructose.

Explain how the products could be tested to show that they had not been contaminated by the enzyme.

(3)

.....

.....

.....

.....

.....

.....

(iii) State two other advantages of using immobilised enzymes.

(2)

1 .....

.....

2 .....

.....

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

3. Question 6 · part a.ii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 7 · part a.iii · 3 marks

November 2020 Paper 2R

Further practice · Attempt part a.iii only.

7 Chymosin is an enzyme found in young humans, but not in adults. It converts soluble proteins in milk into solid proteins.

(a) (i) Suggest the advantages to a young human of having chymosin in their alimentary canal.

(2)

(ii) Explain why chymosin is only needed in young humans.

(2)

(iii) Describe a test to show that the solid formed by the action of chymosin on milk is a protein.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

4. Question 7 · part a.iii · 3 marks

November 2020 Paper 2R

Further practice · Attempt part a.iii only.

(b) A student investigates the effect of carbon dioxide on the activity of chymosin.

The student bubbles different volumes of carbon dioxide gas into five samples of milk.

He then adds chymosin to each sample and records the time taken for the milk protein to become solid.

The student repeats this method three times.

The table shows the student's results.

Number of bubbles of CO_2	Time taken for chymosin to turn milk protein solid, in seconds				
	Test 1	Test 2	Test 3	Test 4	Mean
100	253	257	250	260	255
150	238	232	241	229	
200	216	214	219	211	215
250	208	202	212	198	205
300	210	200	199	311	203

(i) State three factors that the student should control.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(ii) Calculate the missing mean (average) time taken for 150 bubbles.

(2)

mean time taken = ...

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

20

P 6 2 0 5 4 A 0 2 0 2 4

4. Question 7 · part a.iii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 3 · part a.ii · 3 marks

June 2021 Paper 1

Further practice · Attempt part a.ii only.

3 Enzymes are protein molecules.

(a) (i) Which sub-units form enzymes?

(1)

- ☒ **A** amino acids
- ☒ **B** fatty acids
- ☒ **C** glucose
- ☒ **D** glycerol

(ii) Describe how a solution can be tested to see if it contains protein.

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

5. Question 3 · part a.ii · 3 marks

June 2021 Paper 1

Further practice · Attempt part a.ii only.

(b) A scientist investigates the effect of temperature on two different enzymes, A and B.

Enzyme A and enzyme B both digest proteins.

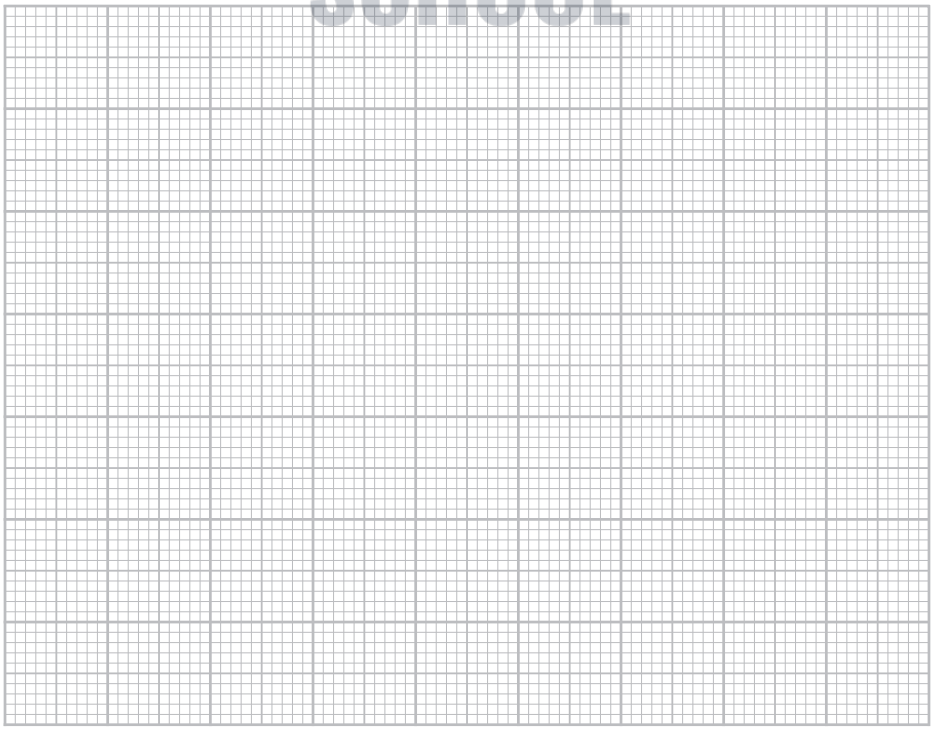
The results of the scientist's investigation are shown in the table.

Temperature in °C	Rate of protein digestion in arbitrary units	
	Enzyme A	Enzyme B
10	0	0
20	5	10
30	5	40
40	20	80
50	60	140
60	80	200
70	20	280
80	0	320
90	0	80

(i) Plot the scientist's results on the grid.

Draw the best fit curve for each enzyme.

(5)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 6 6 4 3 2 A 0 8 2 4

5. Question 3 · part a.ii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 2 · part b · 3 marks

June 2023 Paper 2

Further practice · Attempt part b only.

2 The table gives the nutritional information for a pot of yoghurt.

	Mass per 100 g of yoghurt	Mass per 85 g pot
fat	2.7 g	2.3 g
carbohydrate	12.6 g	10.7 g
protein	6.1 g	
calcium	149 mg	127 mg
vitamin D	0.9 µg	0.8 µg

(a) Calculate the mass of protein in one pot of yoghurt.

Give your answer to two significant figures.

(3)

OLevels.COM — SCHOOL

mass of protein = ... g

(b) Describe how the yoghurt could be tested to show that protein is present.

(3)

.....

.....

.....

.....

.....

.....

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

6. Question 2 · part b · 3 marks

June 2023 Paper 2

Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(c) A pot of yoghurt provides 16% of a person's daily requirement of calcium.

Calculate how many pots of yoghurt a person would need to eat to obtain their daily requirement of calcium.

(2)

number of pots = ...

(d) Explain why yoghurt is good for bone development.

(5)

OLevels.com SCHOOL

(Total for Question 2 = 13 marks)

6. Question 2 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 4 · part b · 2 marks

June 2024 Paper 1R

Further practice · Attempt part b only.

4 (a) The table gives information on how the time taken to digest starch by an enzyme varies at different pH levels.

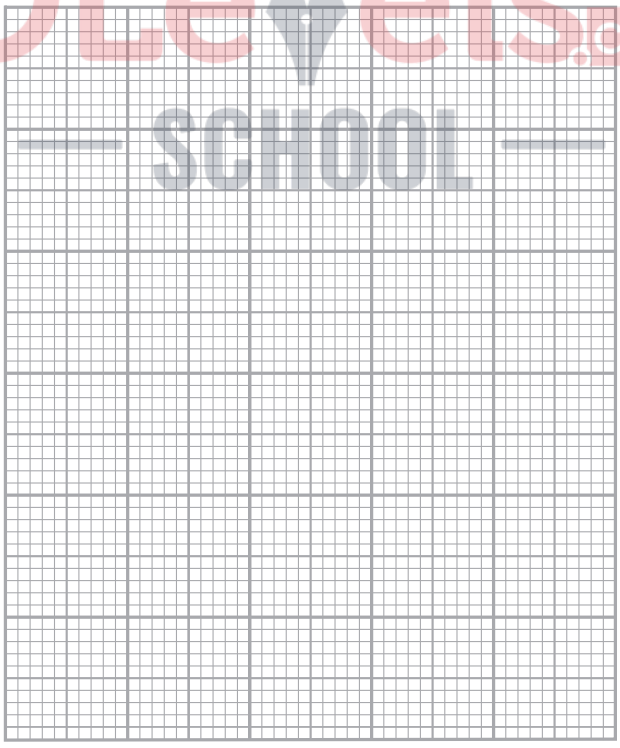
pH	Time taken to digest starch in minutes
2	60
4	45
6	20
8	35
10	55

(i) Plot a graph using the data in the table.

(3)

(ii) Draw a curve of best fit.

(1)



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

7. Question 4 · part b · 2 marks

June 2024 Paper 1R

Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA

(iii) Use the graph to determine the optimum pH for the digestion of starch.

(1)

(iv) Describe the trend shown by the graph.

(2)

(v) Explain why the time taken to digest starch is different at pH 2 compared with the time taken at pH 6.

(3)

OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA

(vi) Name a part of the body where starch digestion takes place.

(1)

DO NOT WRITE IN THIS AREA

(b) Describe a test for starch in a sample of food.

(2)

(Total for Question 4 = 13 marks)

7. Question 4 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 2 · part b.iv · 5 marks

November 2024 Paper 2

Further practice · Attempt part b.iv only.

2 (a) Describe what is meant by a balanced diet.

(2)

.....

.....

.....

.....

(b) A label on a pack of four slices of roast beef gives information about how much of each dietary component is present in the meat.

It also gives the percentage of the recommended daily amount for each dietary component.

Table 1 and table 2 give these values.

Energy in kJ	Fat in g	Sugars in g	Salt in g
470	2.4	less than 0.4	0.6

Table 1

Percentage of recommended daily amount (%)			
energy	**fat**	**sugars**	**salt**
4.0	4.0	less than 1.0	12.0

Table 2

(i) Name another dietary component that could be listed on the pack of meat.

(1)

.....

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 7 5 9 5 9 A 0 4 2 4

8. Question 2 · part b.iv · 5 marks

November 2024 Paper 2

Further practice · Attempt part b.iv only.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(ii) Using data from the tables, calculate the mass of fat, in g, that is needed to achieve the recommended daily amount.

(3)

mass = ... g

(iii) Calculate the number of slices of meat a person would need to eat to supply a recommended daily energy intake of 11 800 kJ.

(2)

OLevels.com SCHOOL

number of slices = ...

P 7 5 9 5 9 A 0 5 2 4

8. Question 2 · part b.iv · 5 marks

November 2024 Paper 2

Further practice · Attempt part b.iv only.

(iv) When meat is rubbed on a piece of paper it leaves a greasy mark due to the presence of fat in the meat.

Describe another test to show the presence of fat in the meat.

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 2 = 13 marks)

8. Question 2 · part b.iv · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 2 · part c.i · 5 marks

June 2019 Paper 1R

Further practice · Attempt part c.i only.

2 Humans need a balanced diet.

(a) Vitamins are needed in a balanced diet.

State four other components of a balanced diet.

(2)

1 .....

2 .....

3 .....

4 .....

(b) Explain why vitamin C is needed in a balanced diet.

(2)

.....

.....

.....

.....

OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

9. Question 2 · part c.i · 5 marks

June 2019 Paper 1R

Further practice · Attempt part c.i only.

(c) The vitamin C content of fruit juices varies.

(i) Design an investigation to compare the vitamin C content of fresh fruit juice and long-life fruit juice.

(5)

(ii) State two variables that should be controlled in this investigation.

(2)

1 .....

2 .....

(Total for Question 2 = 11 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
OLevels.com SCHOOL

9. Question 2 · part c.i · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 7 · part b · 5 marks

June 2022 Paper 2

Further practice · Attempt part b only.

7 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Our body is under constant attack from free radicals. Free radicals are molecules produced during chemical reactions that take place in body cells. Free radicals can change the structure and function of cells and damage genetic material. Some scientific evidence also suggests that free radicals can affect the flow of substances that enter and leave a cell.

Vitamin C is water-soluble and an important component of the daily diet. The recommended intake of vitamin C is 90 mg per day. Vitamin C helps to strengthen the immune system and improves the absorption of iron into the body. Vitamin C is also known as an antioxidant. Antioxidants protect cells from damage by free radicals and other substances by changing the way that free radicals behave.

Water and high temperatures can reduce the vitamin C content of food. For example, 40 g of fresh broccoli provides 80% of the recommended daily intake of vitamin C. The same mass of boiled broccoli provides 60% of the recommended daily intake of vitamin C. Boiling is the biggest cause of vitamin C loss, although baking and frying also lower vitamin C content.

(a) (i) Free radicals can damage genetic material in cells.

Describe one effect that this could have on body cells. (lines 2 and 3)

(2)

(ii) Suggest how antioxidants, such as vitamin C, allow cells to function normally. (lines 5 and 6 and lines 8 to 10)

(2)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

10. Question 7 · part b · 5 marks

June 2022 Paper 2

Further practice · Attempt part b only.

(iii) Calculate the mass, in grams, of boiled broccoli that should be eaten to provide the recommended daily intake of vitamin C. (lines 12 to 14)

(3)

mass = ... g

(b) DCPIP is a chemical used to determine the amount of vitamin C in food.

Describe a method, using DCPIP, to compare the amount of vitamin C in fresh orange juice with that in boiled orange juice.

(5)

(Total for Question 7 = 12 marks)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

10. Question 7 · part b · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 2 · part a · 4 marks

November 2021 Paper 2

Further practice · Attempt part a only.

2 A student uses these three tests to identify the contents of four powders, W, X, Y and Z.

- iodine test
- Benedict's test
- biuret test

The table shows the student's results.

	Colour observed after testing			
Test	**Powder W**	**Powder X**	**Powder Y**	**Powder Z**
iodine	yellow / brown	black	black	yellow / brown
Benedict's	blue	brick red	blue	brick red
biuret	purple	blue	blue	purple

(a) Describe how the student should safely carry out the Benedict's test on the powders.

(4)

OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

11. Question 2 · part a · 4 marks

November 2021 Paper 2

Further practice · Attempt part a only.

DO NOT WRITE IN THIS AREA

(b) (i) Identify which powder gives a positive result for starch but a negative result for the two other tests.

(1)

(ii) Identify which powders contain protein but no starch.

(2)

(iii) Identify which powder contains reducing sugar but no protein.

(1)

(iv) Give the contents of powder W.

(1)

(Total for Question 2 = 9 marks)

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

11. Question 2 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 2 · part a.i · 4 marks

January 2022 Paper 1

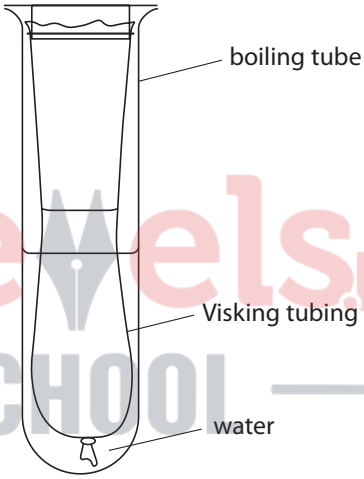
Further practice · Attempt part a.i only.

2 A student investigates absorption in the ileum by using a model gut made of Visking tubing.

This is the student's method.

- tie a knot in one end of a 15 cm length of Visking tubing to make a bag
- place 5 cm^3 of starch suspension and 5 cm^3 of glucose solution into the bag
- rinse the outside of the bag with water
- place the bag into a boiling tube containing distilled water
- leave the bag for one hour

The diagram shows the student's apparatus.



After an hour the contents of the Visking tubing bag are tested for glucose and starch.

The water in the boiling tube is also tested for glucose and starch.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

12. Question 2 · part a.i · 4 marks

January 2022 Paper 1

Further practice · Attempt part a.i only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(a) (i) Describe how the student should test the contents of the Visking tubing bag for glucose.

(4)

(ii) Describe how the student should test the contents of the Visking tubing bag for starch.

(3)

(iii) Explain why the outside of the Visking tubing bag was rinsed with water before being placed in the boiling tube.

(3)

OLevels.com SCHOOL

12. Question 2 · part a.i · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

13. Question 2 · part a.ii · 3 marks

January 2022 Paper 1

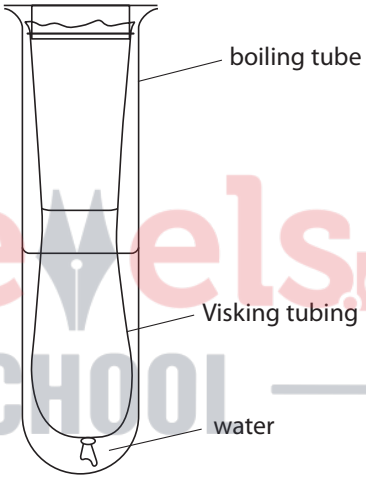
Further practice · Attempt part a.ii only.

2 A student investigates absorption in the ileum by using a model gut made of Visking tubing.

This is the student's method.

- tie a knot in one end of a 15 cm length of Visking tubing to make a bag
- place 5 cm^3 of starch suspension and 5 cm^3 of glucose solution into the bag
- rinse the outside of the bag with water
- place the bag into a boiling tube containing distilled water
- leave the bag for one hour

The diagram shows the student's apparatus.



After an hour the contents of the Visking tubing bag are tested for glucose and starch.

The water in the boiling tube is also tested for glucose and starch.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

13. Question 2 · part a.ii · 3 marks

January 2022 Paper 1

Further practice · Attempt part a.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(a) (i) Describe how the student should test the contents of the Visking tubing bag for glucose.

(4)

(ii) Describe how the student should test the contents of the Visking tubing bag for starch.

(3)

(iii) Explain why the outside of the Visking tubing bag was rinsed with water before being placed in the boiling tube.

(3)

Olevels.com SCHOOL

13. Question 2 · part a.ii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

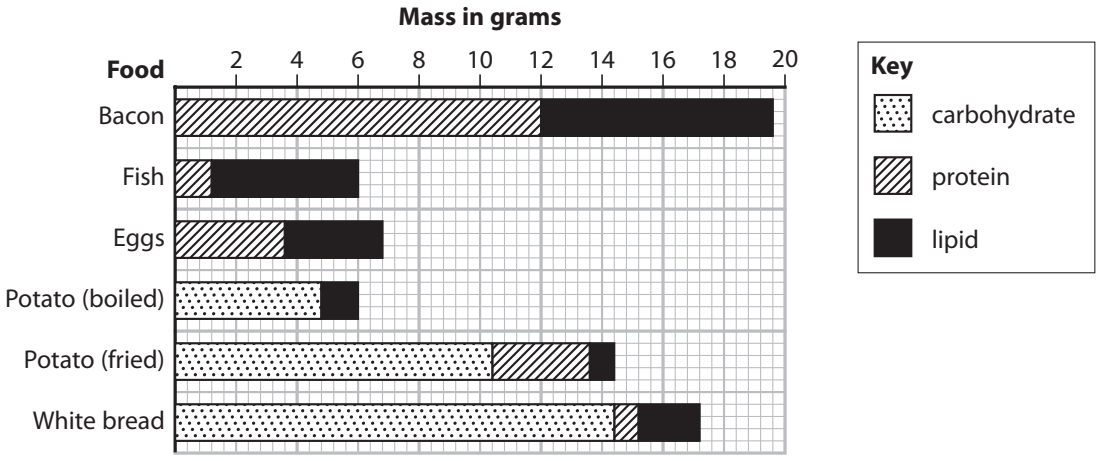
14. Question 7 · part b · 3 marks

January 2022 Paper 1

Further practice · Attempt part b only.

7 Most diets contain carbohydrates, lipids and proteins.

The bar chart shows the masses of each of these in 26 g samples of different foods.



(a) (i) State the names of the three chemical elements found in all carbohydrates, lipids and proteins.

(1)

(ii) State the name of one chemical element found in proteins but not found in carbohydrates or lipids.

(1)

(iii) Deduce the mass of lipid in 26 g of bacon.

(2)

mass of lipid = ... g

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

20

P 6 7 0 5 9 A 0 2 0 2 4

14. Question 7 · part b · 3 marks

January 2022 Paper 1

Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA

(iv) The energy in one gram of protein is 17 kJ and the energy in one gram of lipid is 37 kJ.

Calculate the total energy in 26 g of fish.

(3)

total energy = ... kJ

DO NOT WRITE IN THIS AREA

(b) Describe how to test for the presence of lipid in the fish.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

DO NOT WRITE IN THIS AREA

(Total for Question 7 = 10 marks)

OLevels.COM — SCHOOL —

14. Question 7 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

15. Question 4 · part a.i · 3 marks

June 2022 Paper 1R

Further practice · Attempt part a.i only.

4 (a) Lipids are digested in the small intestine.

(i) Describe a test for lipids in food.

(3)

(ii) Name the enzyme that digests lipids.

(1)

(iii) Name the products of lipid digestion.

(2)

OLevels.COM SCHOOL

(b) A student investigates how the pH of a lipid solution, containing an enzyme that digests lipids, changes over 10 minutes at two different temperatures.

The tables show the student's results.

Temperature at 37 °C

Time in minutes	1	2	3	4	5	6	7	8	9	10
pH	8.5	8.0	7.5	7.2	7.0	6.8	6.2	5.8	5.5	5.5

Temperature at 80 °C

Time in minutes	1	2	3	4	5	6	7	8	9	10
pH	9.2	9.0	8.8	8.8		8.6	8.6	8.4	8.4	8.4

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

15. Question 4 · part a.i · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

16. Question 4 · part c · 4 marks

June 2022 Paper 2

Further practice · Attempt part c only.

4 A student tests foods W, X and Y for the presence of different nutrients.

The student tests each of the foods using three different solutions, A, B and C.

The table shows the student's results.

Food	Solution used in test		
	A	B	C
W	no change	no change	blue-black
X	lilac	no change	no change
Y	no change	brick-red	no change

(a) Name solution C.

(1)

.....

(b) Name the nutrient present in food X.

(1)

.....

(c) Describe a safe method the student could use to test a food using solution B.

(4)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for Question 4 = 6 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

O Levels .COM — SCHOOL —

16. Question 4 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

17. Question 2 · part b · 5 marks

November 2023 Paper 1

Further practice · Attempt part b only.

2 To find out if a person has diabetes, they can be given a glucose tolerance test.

This is how the test is done.

- the person does not eat for 12 hours
- after the 12 hours, blood glucose levels are measured
- the person is then given a solution of glucose to drink
- the blood glucose levels of the person are measured every 30 minutes for 3 hours

The table shows the results for a person without diabetes and a person with diabetes.

Time in minutes	Blood glucose levels in mg per 100 cm^3	
	person without diabetes	person with diabetes
0	65	110
30	120	215
60	130	230
90	95	200
120	80	190
150	65	180
180	65	170

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

17. Question 2 · part b · 5 marks

November 2023 Paper 1

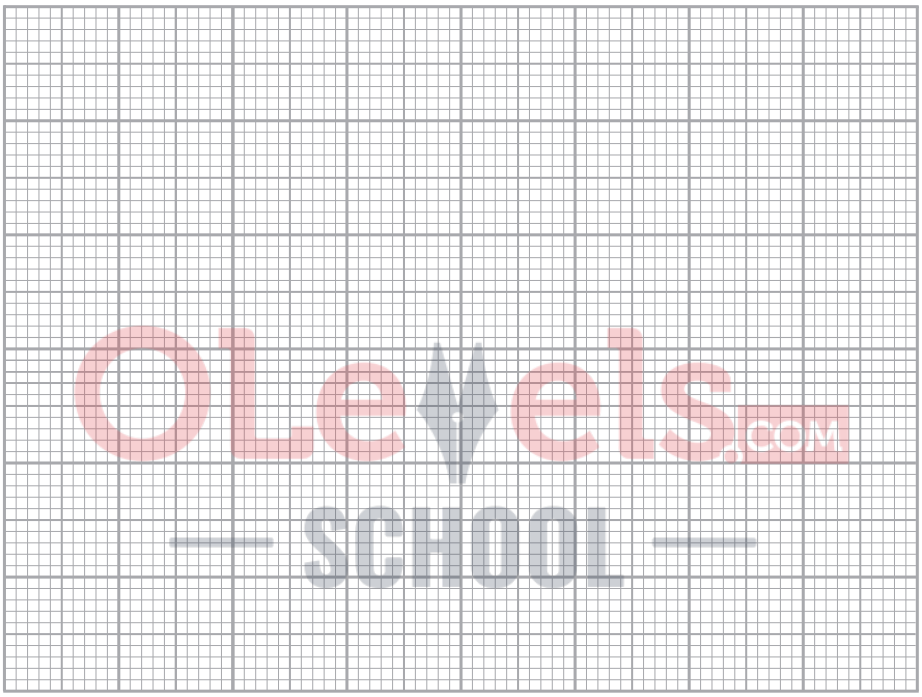
Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(a) (i) Plot the results on the grid.

Join the points with straight lines.

(6)



(ii) Determine the difference between the highest and lowest blood glucose levels for each person.

(2)

without diabetes = ... mg per 100 cm^3

with diabetes = ... mg per 100 cm^3

17. Question 2 · part b · 5 marks

November 2023 Paper 1

Further practice · Attempt part b only.

(iii) Give **three** differences between the glucose levels for the two people during the three hours.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(b) One symptom of diabetes is glucose in the urine.

Describe how a sample of urine could be tested safely for glucose.

(5)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for Question 2 = 16 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THI

OLevels.COM — SCHOOL —

17. Question 2 · part b · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL